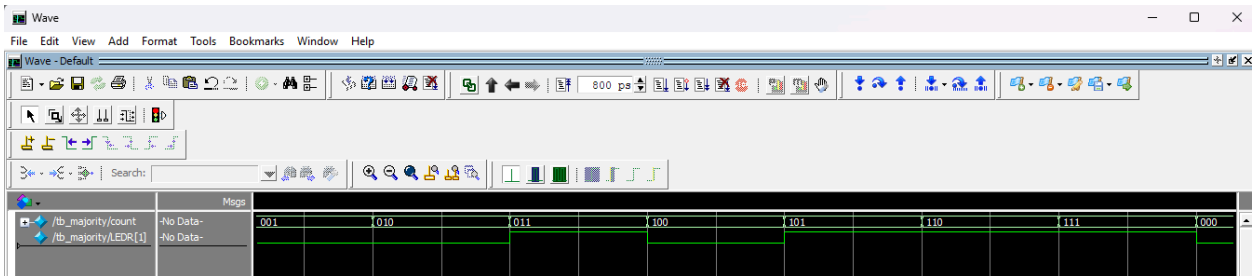
Part 0



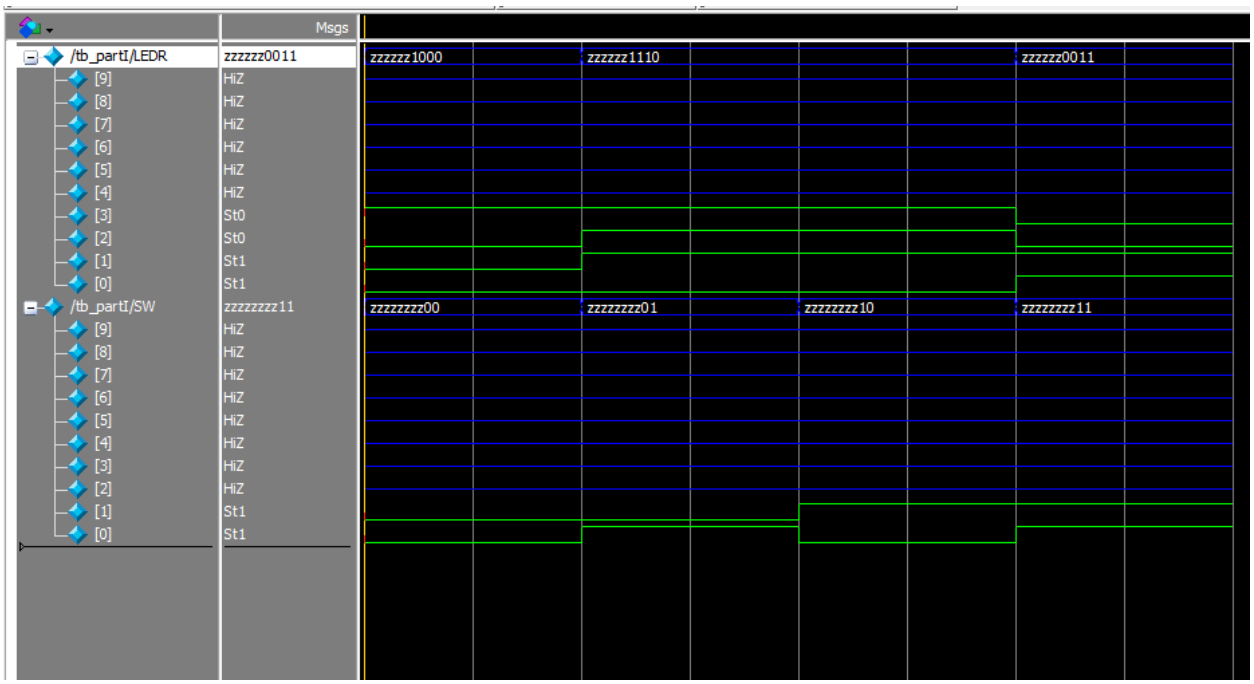
Part I

Probable Output:

Inputs: SW[0], SW[1]

SW[1]	SW[0]	LEDR[0] AND	LEDR[1] OR	LEDR[2] XOR	LEDR[3] NAND
0	0	0	0	0	1
0	1	0	1	1	1
1	0	0	1	1	1
1	1	1	1	0	o

Waveform Simulation:



Part 2

<Truth Table>

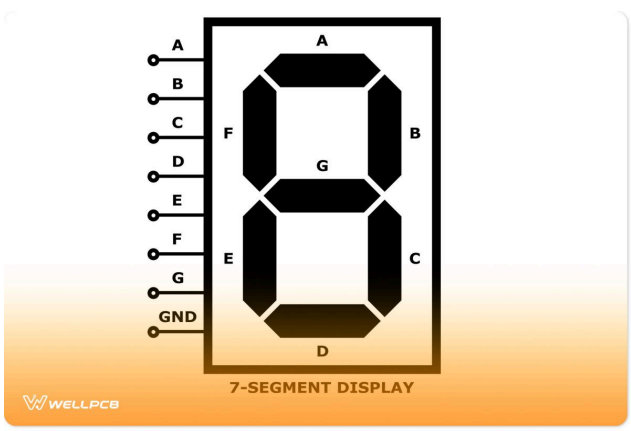
SW[3:0]	Decimal	HEX1	HEX0	HEX1[7:0]	HEX0[7:0]
0000	00	0	0	11000000	11000000
0001	01	0	1	11000000	11111001
0010	02	0	2	11000000	10100100
0011	03	0	3	11000000	10110000
0100	04	0	4	11000000	10011001
0101	05	0	5	11000000	10010010
0110	06	0	6	11000000	10000010
0111	07	0	7	11000000	11111000
1000	08	0	8	11000000	10000000
1001	09	0	9	11000000	10010000
1010	10	1	0	11111001	11000000

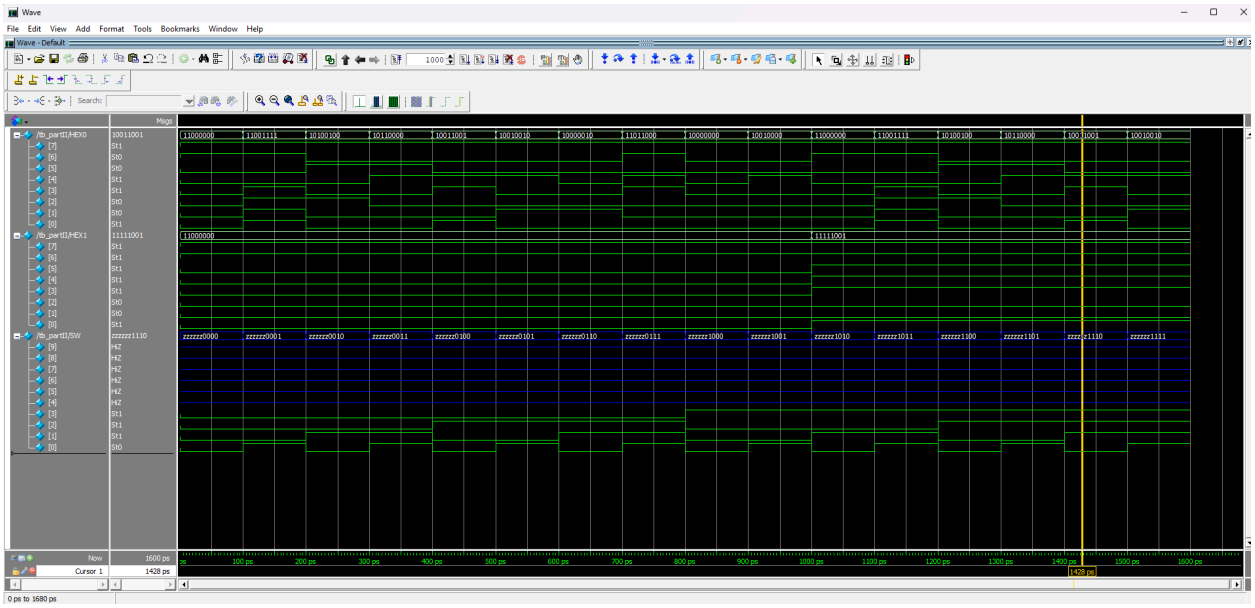
1011	11	1	1	11111001	11111001
1100	12	1	2	11111001	10100100
1101	13	1	3	11111001	10110000
1110	14	1	4	11111001	10011001
1111	15	1	5	11111001	10010010

<Karnaugh Maps & Logic Equations from Ishani>

HEX1[0:3] = SW[3]SW[2] + SW[3]SW[1]
HEX1[4:5] = 0
HEX1[6] = 1
HEX0[0] = [SW3′SW1′(SW2SW0′ + SW2′SW0)] + [SW3SW1(SW2SW0′ + SW2′SW0)]
HEX0[1] = SW3′SW1′SW0 + SW3SW1SW0 + SW3′SW2SW1SW0′
HEX0[2] = SW3SW2SW1′SW0′ + SW3′SW2′SW1′SW0 + SW3′SW2′SW1SW0′ + SW3SW2′SW1SW0
HEX0[3] = SW3′SW1′(SW2SW0′ + SW2′SW0) + SW1SW0(SW3′ SW2 + SW3SW2′) + SW3SW2SW1SW0′
HEX0[4] = SW3′SW2SW1′ + SW3′SW1SW0 + SW3SW1′SW0 + SW3SW2SW1
- Simplifies to SW[0] + SW[1]SW[2]SW[3] + SW[2]SW[1]’SW[3]’

HEX0[5] = SW3SW2SW1′ + SW3′SW2′SW1
HEX0[6] =SW3′SW2′SW1′ + SW3SW2′SW1 + SW3′SW2SW1SW0




Wave
File Edit View Add Format Tools Bookmarks Window Help
Wave - Default
/tb_part1/HEX0
/tb_part1/HEX1
/tb_part1/SW
Now
1600 ps
Cursor 1
1428 ps
0 ps to 1680 ps